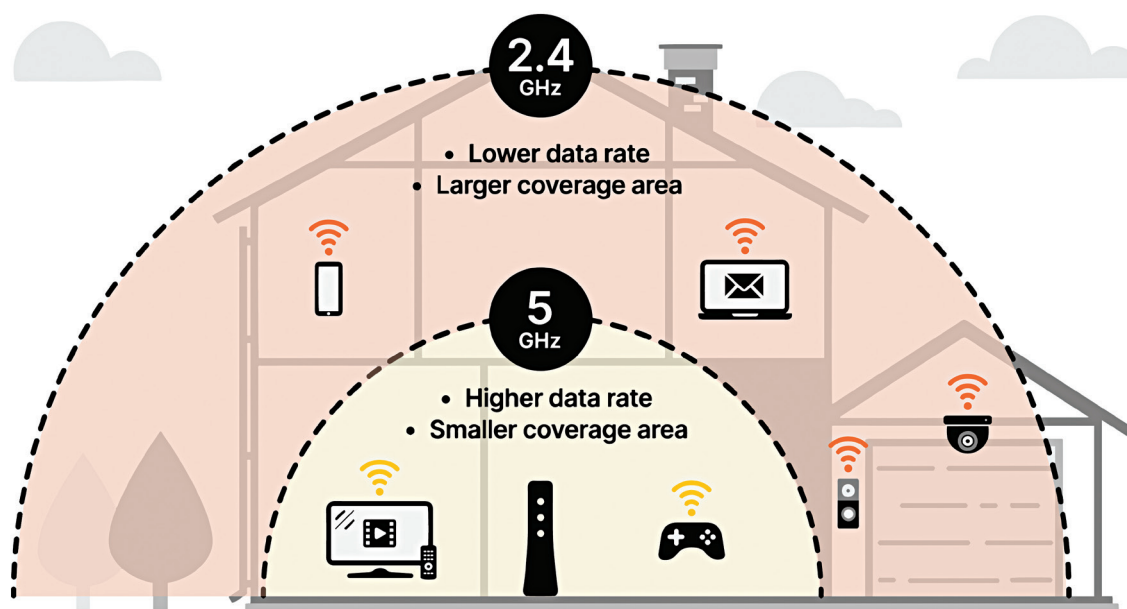




3. If you have many 2.4-only devices, plan an IoT lane that's 2.4-first (Section 7).

Rule of thumb:

- Humans (laptops/phones/TVs) → 5/6 GHz where possible.
- Smart home basics (plugs/bulbs/sensors) → stable 2.4 GHz is usually fine.

2) Do a dead-zone test you can trust (30 min)

You're not chasing maximum speed. You're checking whether devices can stay connected where they live.

Step-by-step: build a quick "coverage map"

1. Sketch your floor plan: rooms, corridor, router location.
2. Pick 8–12 test points: corners, near the main door, near the TV unit, near the kitchen, and any place you want sensors/cameras.
3. At each point, record:
 - which band you're on (2.4 or 5)
 - a quick speed test (download is enough)
 - one real-world check: stream a short video or run a voice query
4. Mark each point:
 - Green: reliable
 - Amber: usable but shaky
 - Red: drops/feels broken

How to interpret your results:

- If you have Red where you plan cameras, entry sensors, or motion sensors, you fix coverage first.
- If you have Amber in bedrooms, you may "feel" it as slow voice responses and delayed routines.
- A smart home doesn't need gigabit everywhere. It needs predictable connectivity.

3) Router placement is key

Routers often end up where the line enters: near the door, behind the TV, inside a cabinet. Great for aesthetics. Bad for radio waves.

Step-by-step: place the main router like a "center speaker"

1. Go as central as you can (avoid corner rooms).
2. Elevate it (shelf/table beats the floor).
3. Keep it open (no metal cabinets, no sealed TV units).
4. Keep it away from noisy neighbors (microwave, thick beams).



If you can't move the router because the line is fixed, mesh (or a wired access point) is the grown-up solution.

4) Tune the basics: channels and band steering (10 min)

Even with good placement, bad "auto" decisions can cause flakiness – especially on 2.4 GHz in apartment buildings.